

FOURTH SEMESTER DIPLOMA EXAMINATION IN BIO MEDICAL ENGINEERING

ANALYTICAL AND DIAGNOSTIC EQUIPMENTS

(Maximum Marks : 100)

TIME : 3HRS

PART - A

Answer all Questions in one or two sentences. Each question carries **2 marks**.

I.

- 1) Define resolution?
- 2) List any two types of monochromators?
- 3) Define systolic blood pressure?
- 4) Name any two electrical isolation techniques used in biomedical devices?
- 5) List any two advantages of digital meters over analog meters?

5x2=10 marks

PART - B

Answer any **FIVE** of the following questions. Each question carries **6 marks**.

II.

- 1) Describe the working of a galvanometer with a neat labeled sketch?
- 2) Explain the constructional details of catheter tip electrode for the measurement of PO₂ ?
- 3) Summarize the working of any two transducers used for temperature measurement?
- 4) Interpret any six application of EEG in medical field?
- 5) Identify the basic blocks present in bio signal sensing and recording system?
- 6) Differentiate between invasive and non invasive methods of BP measurement?
- 7) Explain hemoglobin meter with block diagram?

5x6=30 marks

PART - C

Answer one full question from each Unit. Each question carries 15 marks.

UNIT I

- III** a) Summarize the working of a CRT with neat diagram? **8**
b) Explain the working of circular chart recorder with a sketch? **7**

OR

- IV.** a) Describe the AC voltage measurement using digital multimeter? **8**
b) Explain the working of XY recorder with a labeled neat sketch? **7**

UNIT II

- V.** a) Explain the electrode used for pH measurement with a labeled sketch? **8**
b) Summarize the principle and working of colorimeter? **7**

OR

- VI a)** Explain the principle and construction of electrode for the measurement of PCO₂? 10
- b) Derive the relationship between energy absorption and the concentration of absorbent? 5

UNIT III

- VII a)** Describe the working of coulter blood cell counter with labeled diagram? 10
- B) Distinguish between Arterial BP and Ventricular BP? 5

OR

- VIII a)** Explain the principle and working of ultrasound blood flow meter? 10
- b) Describe the BP measurement using sphygmomanometer? 5

UNIT IV

- IX a)** Illustrate the following ECG lead configurations with neat sketch?
- (i) Bipolar lead configuration 4
- (ii) Augmented lead configuration 4
- (iii) Unipolar lead configuration 2
- b) Explain vector cardiography? 5

OR

- X a)** Summarize the 10-20 electrode placement for EEG recording? 8
- b) Explain the characteristics and requirements of amplifiers used in biomedical devices? 7

15x4=60

